Chapter 15 — Testing and Debugging

What you will learn. Why automated tests pay for themselves, and how to write them: the built-in `unittest` style, then **pytest** — plain `assert`, fixtures, `parametrize`, `tmp_path`, marks and `conftest.py`. How to isolate code from slow or external dependencies with **test doubles** and `unittest.mock` (`Mock`, `patch`, `side_effect`, autospec). How to measure what your tests exercise with **coverage**, the red-green-refactor **TDD** loop, executable examples with **doctest**, and **property-based testing** with Hypothesis. Then debugging: reading tracebacks, `pdb` and `breakpoint()`, logging instead of `print`, and a checklist of the bugs Python programmers hit most.

Prerequisites. Functions, classes, exceptions (Chapters 7, 9, 10). Logging basics are in Chapter 14.

Code. All code is in `chapters/15-testing-and-debugging/code/`: a small module `inventory.py` plus its tests. Install the test tools once, then run everything from that folder:

```
pip install pytest pytest-cov hypothesis
cd chapters/15-testing-and-debugging/code
pytest -q                       # 31 passed, 1 skipped, 1 xfailed
python debugging_demo.py
```



15.1 Why write tests?

What it is

An **automated test** is code that runs your code with known inputs and checks the result. Vocabulary used in this chapter:

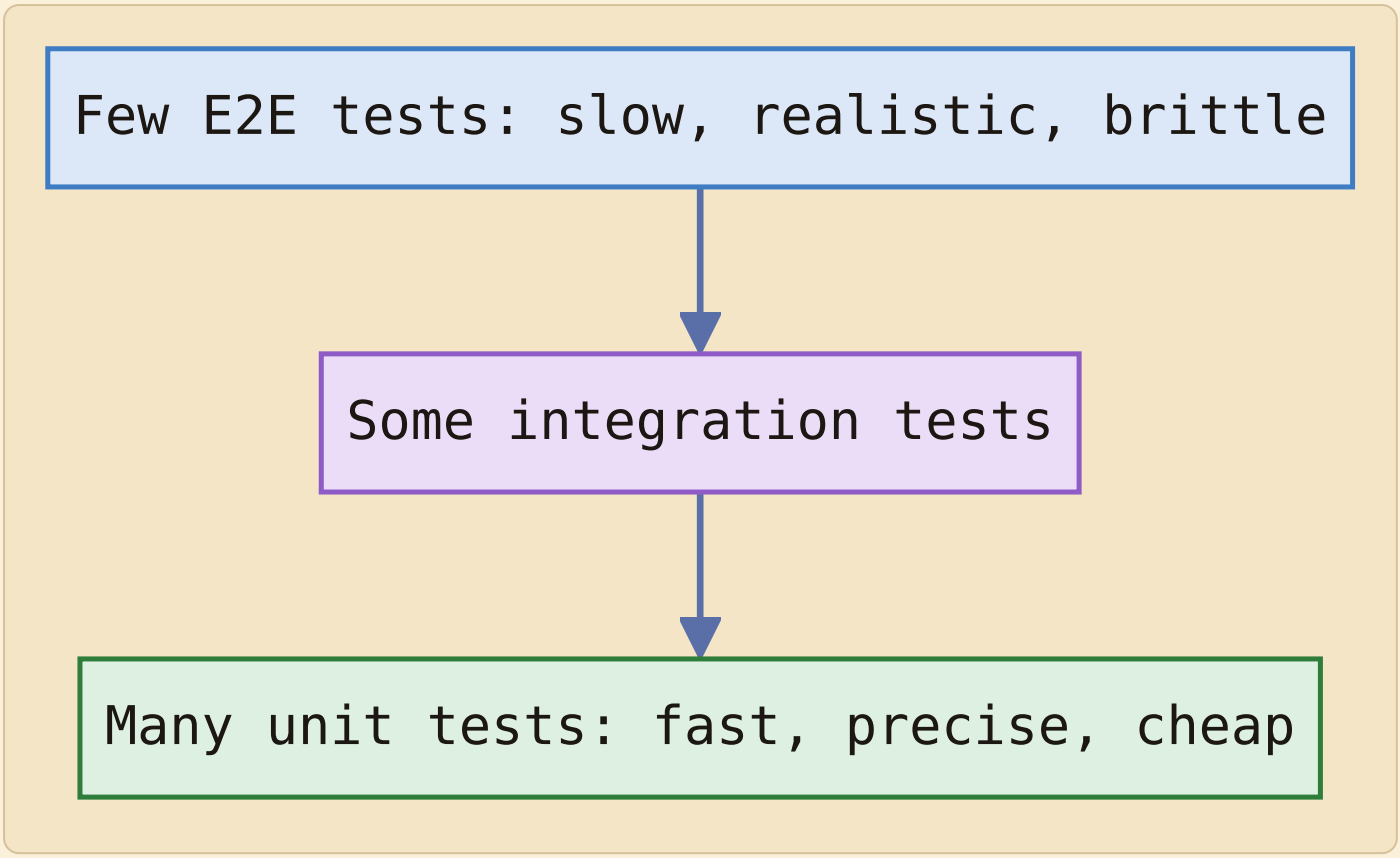
Term	Meaning
Unit test	tests one small piece (a function, a class) in isolation, in milliseconds
Integration test	tests several pieces together (your code + a real database or file system)
End-to-end (E2E) test	drives the whole system like a user (HTTP calls, browser)
Test suite	all the tests of a project
Regression	a bug that comes back, or a change that breaks something that used to work
SUT	*system under test* — the code being tested
Arrange–Act–Assert	the usual three-part shape of a test: set up, do the thing, check the outcome

Why it matters

1. **Confidence to change code.** Refactoring without tests is guessing. With a good suite, you change code and learn in seconds whether anything broke.
2. **Bugs are cheaper earlier.** A failing test on your laptop costs minutes; the same bug in production costs incidents, data fixes and trust.
3. **Tests are executable documentation** — they show exactly how the code is meant to be used.

4. **Better design.** Code that is hard to test is usually too coupled; writing tests pushes you toward small functions with explicit inputs.
5. In Python specifically, there is **no compiler** to catch a typo in a rarely-run branch — only running the code finds it (type checkers help, Chapter 13).

How it works — the test pyramid



Most tests should be fast unit tests; fewer integration tests; a handful of end-to-end tests for the critical paths. Tests run automatically in **CI** (continuous integration — e.g. GitHub Actions runs `pytest` on every push/pull request).

What makes a good test

- **Fast** (milliseconds), **isolated** (no dependence on other tests or their order), **repeatable** (same result every time — no real clock, network, or unseeded randomness), **self-checking** (pass/fail, no human reading output), and **focused** (one behaviour; the name says which).
- Test **behaviour, not implementation**: check what a function returns or changes, not which private helpers it calls. Then refactoring doesn't break tests.

Real-world examples

CPython's own suite (`python -m test`) is written with `unittest`; NumPy, pandas, pip and most modern projects use pytest; Django ships a `unittest`-based runner. Pull requests on these projects are merged only when CI is green.

TL;DR

Tests let you change code safely. Many fast unit tests, fewer integration and E2E tests, all run in CI. Test behaviour, keep tests isolated and deterministic.

15.2 The module under test

Every example tests the same small module, `inventory.py` (abridged — `# ...` marks omitted lines):

```python
class OutOfStock(Exception):
    """Raised when removing more units than are available."""

def slugify(name: str) -> str:
    """Lower-case, keep letters/digits, join words with '-'.

    >>> slugify("Hello, World!")
    'hello-world'
    """
    words = re.findall(r"[a-z0-9]+", name.lower())
    return "-".join(words)

def apply_discount(price: float, percent: float) -> float:
    if not 0 <= percent <= 100:
        raise ValueError(f"percent must be 0-100, got {percent}")
    return round(price * (1 - percent / 100), 2)

@dataclass
class Inventory:
    stock: dict[str, int] = field(default_factory=dict)

    def add(self, sku: str, qty: int) -> None:
        if qty <= 0:
            raise ValueError("qty must be positive")
        self.stock[sku] = self.stock.get(sku, 0) + qty

    def remove(self, sku: str, qty: int) -> None:
        have = self.stock.get(sku, 0)
        if qty > have:
            raise OutOfStock(f"{sku}: want {qty}, have {have}")
        self.stock[sku] = have - qty

    def low_stock(self, threshold: int) -> list[str]: ...   # built with TDD in §15.8
    def total_units(self) -> int: ...
    def save(self, path: Path) -> None:
        path.write_text(json.dumps(self.stock, sort_keys=True))
    @classmethod
    def load(cls, path: Path) -> "Inventory": ...

def price_in(currency: str, usd: float, client: RateClient) -> float:
    """Convert a USD price using an external rate service (mocked in tests)."""
    rate = client.get_rate(currency)
    if rate <= 0:
        raise ValueError("bad rate")
    return round(usd * rate, 2)
```

Note the design choice in `price_in`: the rate service is **passed in** (dependency injection) instead of created inside. That single decision makes it trivial to test with a fake (§15.5).

15.3 `unittest` basics

What it is

`unittest` is the standard library's test framework, modelled on Java's JUnit (the "xUnit" family). Tests are methods of a class that inherits `unittest.TestCase`.

How it works

1. Create `class TestX(unittest.TestCase)`.
2. Each method whose name starts with `test` is one test.
3. `setUp()` runs before **each** test, `tearDown()` after each; `setUpClass` / `tearDownClass` (class methods) run once per class.
4. Check results with assertion methods: `assertEqual`, `assertTrue`, `assertIn`, `assertIsNone`, `assertAlmostEqual` (floats, 7 decimal places by default), `assertRaises` (as a context manager), `assertLogs`.
5. Skip with `@unittest.skip(reason)` / `skipIf(cond, reason)`; `@unittest.expectedFailure` for known bugs.
6. Run: `python -m unittest` (auto-discovers `test*.py` files), `python -m unittest -v test_inventory_unittest`, or `unittest.main()` inside the file.

```
class TestInventory(unittest.TestCase):
    def setUp(self):                      # runs before EACH test method
        self.inv = Inventory()
        self.inv.add("mug", 5)

    def test_add_accumulates(self):
        self.inv.add("mug", 3)
        self.assertEqual(self.inv.stock["mug"], 8)

    def test_remove_too_many_raises(self):
        with self.assertRaises(OutOfStock):
            self.inv.remove("mug", 99)

    def test_discount(self):
        self.assertAlmostEqual(apply_discount(19.99, 10), 17.99)

    @unittest.skip("demo of skipping")
    def test_skipped(self):
        self.fail("never runs")
```

Output of `python test_inventory_unittest.py`:

```
test_add_accumulates (__main__.TestInventory.test_add_accumulates) ... ok
test_discount (__main__.TestInventory.test_discount) ... ok
test_remove_too_many_raises (__main__.TestInventory.test_remove_too_many_raises) ... ok
test_skipped (__main__.TestInventory.test_skipped) ... skipped 'demo of skipping'

----------------------------------------------------------------------
Ran 4 tests in 0.000s

OK (skipped=1)
```

Trade-offs — unittest vs pytest

	unittest	pytest
Install	built in	`pip install pytest`
Test shape	class + `self.assertX` methods	plain functions + `assert`
Setup	`setUp` / `tearDown`	fixtures (reusable, composable, scoped)
Data-driven tests	`self.subTest(...)` loop	`@pytest.mark.parametrize`
Failure messages	from the assert method	rewritten `assert` shows values and diffs
Plugins	few	large ecosystem (coverage, xdist, asyncio, django…)
Runs the other's tests	—	yes, pytest runs `unittest` tests unchanged

When to use it

When you can't add dependencies (stdlib-only tools, CPython itself), or in an existing unittest codebase. For everything else, pytest is the community default — and since pytest runs unittest tests, you can migrate gradually.

TL;DR

`unittest`: `TestCase` classes, `test_` methods, `setUp`, `assertEqual` / `assertRaises`; run with `python -m unittest`. Built in, more verbose than pytest.

15.4 pytest

What it is

pytest is the most widely used Python test framework. Tests are plain functions using plain `assert`; pytest discovers them, runs them, and explains failures in detail.

How it works — discovery and running

1. pytest collects files named `test_*.py` or `*_test.py`, functions named `test_*`, and classes named `Test*` (without an `__init__`).
2. Common commands:

Command	Does
`pytest`	run everything under the current folder
`pytest -q` / `-v`	quieter / one line per test
`pytest test_x.py::test_name`	one test
`pytest -k "discount and not bad"`	tests whose names match the expression
`pytest -m "not slow"`	select by mark
`pytest -x` / `--maxfail=2`	stop after the first/second failure
`pytest --lf`	rerun only last-failed tests
`pytest -s`	show `print` output (normally captured)
`pytest --pdb`	open the debugger at a failure
`pytest --durations=5`	list the slowest tests

3. Settings live in `pyproject.toml` under `[tool.pytest.ini_options]` (e.g. `testpaths = ["tests"]`, `markers = ["slow: ..."]`, `addopts = "-q"`).

15.4.1 Plain `assert` and assertion rewriting

pytest rewrites `assert` statements in test files so that on failure it prints the values involved. You write `assert result == expected`, and get a diff for free (see §15.8 for a real one). Helpers:

- `pytest.raises(Exc, match="regex")` — the block must raise `Exc` with a message matching the regex.
- `pytest.approx(x)` — float comparison with tolerance (relative 1e-6 by default): `0.1 + 0.2 == pytest.approx(0.3)`.
- `pytest.warns(Warning)` — the block must emit a warning.

15.4.2 Fixtures

A **fixture** is a function that prepares something a test needs (an object, a temp file, a database connection) and optionally cleans it up afterwards. A test requests a fixture simply by naming it as a parameter — pytest calls the fixture and passes in the result (**dependency injection**).

```
# conftest.py
@pytest.fixture
def inv():
    """A fresh inventory with two products for every test that asks for it."""
    inventory = Inventory()
    inventory.add("mug", 5)
    inventory.add("pen", 10)
    return inventory

# test_inventory_pytest.py
def test_total_units(inv):
    assert inv.total_units() == 15

def test_remove_reduces_stock(inv):
    inv.remove("pen", 4)
    assert inv.stock == {"mug": 5, "pen": 6}

def test_remove_too_many_raises(inv):
    with pytest.raises(OutOfStock, match="want 6, have 5"):
        inv.remove("mug", 6)
```

Rules:

1. **Fresh per test by default** (`scope="function"`): `test_remove_reduces_stock` modifies its `inv`, and the next test still gets the original — tests stay isolated.
2. **Scopes**: `function` (default), `class`, `module`, `package`, `session`. Use wider scopes for expensive, read-only resources (e.g. one database engine per session).
3. **Teardown with `yield`**: code after `yield` runs after the test, even if it fails:

```
@pytest.fixture
def audit_log(tmp_path):
    f = (tmp_path / "audit.txt").open("w")
    yield f          # the test runs here
    f.close()        # cleanup
```

4. Fixtures can use other fixtures (`audit_log` uses `tmp_path`).
5. `autouse=True` applies a fixture to every test in its scope without naming it — use sparingly.
6. Useful **built-in fixtures**: `tmp_path` (a unique temporary directory as a `Path`), `monkeypatch` (temporarily set attributes, env vars, dict items; undone automatically), `capsys` (capture stdout/stderr), `caplog` (capture log records), `request` (info about the requesting test).

15.4.3 `tmp_path` — real files, safely

```
def test_save_and_load_roundtrip(inv, tmp_path):
    path = tmp_path / "stock.json"          # unique temp dir per test
    inv.save(path)
    assert Inventory.load(path) == inv
    assert path.read_text() == '{"mug": 5, "pen": 10}'
```

Each test gets its own directory, so tests never collide and never write into your project. This is often better than mocking the file system: it tests the real code path and is still fast.

15.4.4 `parametrize` — one test, many cases

```
@pytest.mark.parametrize(
    ("price", "percent", "expected"),
    [(100, 0, 100), (100, 25, 75), (19.99, 10, 17.99), (50, 100, 0)],
)
def test_discount(price, percent, expected):
    assert apply_discount(price, percent) == pytest.approx(expected)

@pytest.mark.parametrize(("raw", "slug"), [
    ("Blue Mug", "blue-mug"),
    ("  --Hello__World-- ", "hello-world"),
    ("", ""),
], ids=["simple", "punctuation", "empty"])
def test_slugify(raw, slug):
    assert slugify(raw) == slug
```

Each tuple becomes a **separate test** with its own ID, so one failing case doesn't hide the others. Include edge cases: empty input, zero, boundaries (0 and 100), negative numbers, Unicode.

15.4.5 Marks

A **mark** is a label on a test that changes how it's selected or run.

Mark	Effect
`@pytest.mark.skip(reason=...)`	never run
`@pytest.mark.skipif(sys.platform == "win32", reason=...)`	run only when the condition is false
`@pytest.mark.xfail(reason=..., strict=True)`	expected to fail; with `strict=True`, an unexpected *pass* is reported as a failure, so you notice when the bug is fixed
`@pytest.mark.parametrize(...)`	data-driven cases
`@pytest.mark.slow` (custom)	a label you select with `-m`; register it to avoid "unknown mark" warnings

```
@pytest.mark.slow
def test_slow_path():
    time.sleep(0.2)
    assert True

@pytest.mark.xfail(reason="float rounding: 2.675 is stored as 2.67499...; use Decimal",
                   strict=True)
def test_known_bug():
    assert apply_discount(2.675, 0) == 2.68
```

The xfail documents a genuine float issue: the literal `2.675` is stored in binary as slightly less than 2.675, so `round(2.675, 2)` gives `2.67`. The fix would be `decimal.Decimal`.

15.4.6 `conftest.py`

`conftest.py` is a special file that pytest loads automatically. Fixtures and hooks defined there are available to every test in that directory and below — **no import needed**. Nested folders can have their own `conftest.py`. Ours defines the `inv` fixture and registers the `slow` marker:

```
def pytest_configure(config):
    config.addinivalue_line("markers", "slow: marks a test as slow")
```

(In a real project you'd register markers in `pyproject.toml` instead.)

Output

`pytest -v test_inventory_pytest.py`:

```
test_inventory_pytest.py::test_total_units PASSED                        [  6%]
test_inventory_pytest.py::test_remove_reduces_stock PASSED               [ 12%]
test_inventory_pytest.py::test_remove_too_many_raises PASSED             [ 18%]
test_inventory_pytest.py::test_discount[100-0-100] PASSED                [ 25%]
test_inventory_pytest.py::test_discount[100-25-75] PASSED                [ 31%]
test_inventory_pytest.py::test_discount[19.99-10-17.99] PASSED           [ 37%]
test_inventory_pytest.py::test_discount[50-100-0] PASSED                 [ 43%]
test_inventory_pytest.py::test_discount_rejects_bad_percent[-1] PASSED   [ 50%]
test_inventory_pytest.py::test_discount_rejects_bad_percent[101] PASSED  [ 56%]
test_inventory_pytest.py::test_slugify[simple] PASSED                    [ 62%]
test_inventory_pytest.py::test_slugify[punctuation] PASSED               [ 68%]
test_inventory_pytest.py::test_slugify[empty] PASSED                     [ 75%]
test_inventory_pytest.py::test_save_and_load_roundtrip PASSED            [ 81%]
test_inventory_pytest.py::test_audit_written PASSED                      [ 87%]
test_inventory_pytest.py::test_slow_path PASSED                          [ 93%]
test_inventory_pytest.py::test_known_bug XFAIL (float rounding: 2.67...) [100%]

======================== 15 passed, 1 xfailed in 0.31s =========================
```

`pytest -q -m "not slow" test_inventory_pytest.py` → `14 passed, 1 deselected, 1 xfailed`. `pytest --durations=3` (whole folder):

```
0.20s call     test_inventory_pytest.py::test_slow_path
0.12s call     test_properties.py::test_stock_never_negative
0.04s call     test_properties.py::test_slugify_is_idempotent
```

Common mistakes

- Tests that depend on each other's side effects or order (use fresh fixtures).
- Asserting on floats with `==` (use `pytest.approx`).
- `pytest.raises` wrapping too much code — the exception might come from the setup line instead of the call you meant.
- Module-level mutable state shared between tests.
- Test files with the same basename in different folders without `__init__.py` — pytest reports an import-mismatch error.

In the interview

"I use pytest: plain asserts, fixtures for setup with `yield` for teardown, `parametrize` for edge cases, `tmp_path` and `monkeypatch` for files and environment, and marks to separate slow tests. Shared fixtures go in `conftest.py`."

- *Fixture scope?* Default is per function for isolation; session scope for expensive read-only resources.
- *xfail vs skip?* Skip = don't run; xfail = run, expect failure (strict flags an unexpected pass).

Self-check

1. How does a test get a fixture? → By naming it as a parameter.
2. Where does teardown code go? → After `yield` in the fixture.
3. What does `parametrize` with four tuples produce? → Four separate tests.
4. Does a test need to import fixtures from `conftest.py`? → No, they're discovered automatically.

TL;DR

pytest = functions + `assert`. Fixtures (scoped, `yield` teardown, in `conftest.py`), `parametrize` for cases, `tmp_path` / `monkeypatch` / `capsys` / `caplog` built-ins, marks for skip/xfail/selection.

15.5 Test doubles and `unittest.mock`

What it is

A **test double** is a stand-in for a real dependency (a network API, a payment gateway, the clock) during a test — like a stunt double in a film. The standard names (from Gerard Meszaros' *xUnit Test Patterns*, popularised by Martin Fowler):

Double	What it does	Example
Dummy	passed but never used	`None` for an unused logger argument
Stub	returns canned answers	`get_rate` always returns 83.5
Spy	records how it was called, checked afterwards	a list that collects sent emails
Mock	pre-programmed with expectations about calls, which the test verifies	"`get_rate` must be called once with `'INR'`"
Fake	a working, simplified implementation	an in-memory dict instead of a database

`unittest.mock.Mock` can play stub, spy and mock roles at once.

Why it matters

Real dependencies make tests **slow** (network), **flaky** (service down), **costly** (paid APIs) or **dangerous** (sending real emails, charging cards). They also make some cases hard to reach — how do you make a real service time out on demand?

How it works — `Mock`

1. `m = Mock()` — any attribute access returns another `Mock`, and calling it records the call.
2. `m.get_rate.return_value = 83.5` — **stub** the result.
3. `m.get_rate.side_effect = ...` — dynamic behaviour: an **exception** (raised when called), an **iterable** (one value per call), or a **function** (called with the same arguments).
4. Verify: `assert_called_once_with(...)`, `assert_called_with`, `assert_not_called`, `call_count`, `call_args`, `call_args_list`.
5. `MagicMock` also supports magic methods (`__len__`, `__iter__`, `__enter__`…); `AsyncMock` (3.8+) for `async def` functions.
6. **Autospec** — `create_autospec(RealClass, instance=True)` or `patch(..., autospec=True)` builds a mock that only has the real object's attributes and signatures. A plain `Mock` happily accepts misspelled methods, so a test can pass while the real code would crash.

```python
class FakeRates:                              # hand-written fake
    def __init__(self, rates): self.rates = rates
    def get_rate(self, currency): return self.rates[currency]

def test_with_fake():
    assert price_in("EUR", 10, FakeRates({"EUR": 0.9})) == 9.0

def test_with_mock_stub_and_verify():
    client = Mock()
    client.get_rate.return_value = 83.5          # stub the answer
    assert price_in("INR", 2, client) == 167.0
    client.get_rate.assert_called_once_with("INR")   # verify the interaction

def test_side_effect_exception():
    client = Mock()
    client.get_rate.side_effect = TimeoutError("rate service down")
    with pytest.raises(TimeoutError):
        price_in("EUR", 10, client)

def test_side_effect_sequence():
    client = Mock()
    client.get_rate.side_effect = [0.9, -1.0]     # one value per call
    assert price_in("EUR", 10, client) == 9.0
    with pytest.raises(ValueError, match="bad rate"):
        price_in("EUR", 10, client)
    assert client.get_rate.call_args_list == [call("EUR"), call("EUR")]

def test_autospec_catches_typos():
    client = create_autospec(FakeRates, instance=True)
    with pytest.raises(AttributeError):
        client.get_rates("EUR")                   # misspelled: plain Mock would allow it
```

How it works — `patch`

When code creates or imports its dependency itself (instead of receiving it), use `patch` to swap a name **temporarily**:

```python
def test_patch_where_it_is_used(tmp_path):
    # inventory.py did `import json`; patch the name inventory.json.dumps
    with patch("inventory.json.dumps", return_value="{}") as fake_dumps:
        inventory.Inventory({"a": 1}).save(tmp_path / "x.json")
    fake_dumps.assert_called_once()
    assert (tmp_path / "x.json").read_text() == "{}"
```

1. `patch("module.name")` replaces the attribute for the duration of a `with` block, a decorated test function, or between `start()` / `stop()`; the original is always restored.
2. **Patch where the name is looked up, not where it is defined.** If `shop.py` does `from requests import get`, patch `"shop.get"` — patching `"requests.get"` won't affect the copy of the name already bound in `shop`. (In the example, `inventory.json` *is* the `json` module, so this patch actually affects `json.dumps` everywhere during the block — one reason to keep patches short.)
3. `patch.object(obj, "attr")` patches an attribute of an object you have; `patch.dict(os.environ, {...})` patches dictionary entries.
4. pytest's `monkeypatch` does the same for simple cases: `monkeypatch.setenv("APP_ENV", "test")`, `monkeypatch.setattr(module, "name", value)`.

`pytest -v test_mocking.py`:

```
test_mocking.py::test_with_fake PASSED                                  [ 14%]
test_mocking.py::test_with_mock_stub_and_verify PASSED                  [ 28%]
test_mocking.py::test_side_effect_exception PASSED                      [ 42%]
test_mocking.py::test_side_effect_sequence PASSED                       [ 57%]
test_mocking.py::test_autospec_catches_typos PASSED                     [ 71%]
test_mocking.py::test_patch_where_it_is_used PASSED                     [ 85%]
test_mocking.py::test_monkeypatch_env PASSED                            [100%]
```

Trade-offs

Approach	Pros	Cons
Real dependency (tmp_path, SQLite in memory, local server)	tests the real thing	slower, needs setup
Fake	realistic behaviour, reusable, readable	you must write and maintain it
Mock/stub	quick, can simulate errors/timeouts	couples tests to call details; can drift from the real API
`patch`	works without changing the code	fragile if imports move; hints that dependency injection would be cleaner

When not to mock

- Don't mock what you own and can run cheaply (pure functions, dataclasses, files in `tmp_path`, in-memory SQLite).
- Don't mock the thing you are testing.
- Don't assert every internal call — you'll lock in the implementation. Mock at the **boundary** (network, clock, randomness, third-party SDKs), and prefer checking outcomes over interactions.

Common mistakes

- Patching the wrong path (where defined instead of where used).
- Plain `Mock()` hiding typos — use `autospec`/`spec`.
- Misspelled assertions such as `m.called_once_with(...)` or `m.assret_called()` — on old Pythons some silently "passed" (they just created a child mock). Mock now raises `AttributeError` for names that look like assertions but aren't (typo prefixes like `assret_` since 3.5, and assertion names missing the `assert_` prefix since 3.12), unless created with `unsafe=True`.
- Tests that only prove the mock returns what you told it to.

Real-world examples

Libraries exist for common boundaries: `responses` / `respx` fake HTTP for requests / httpx, `freezegun` / `time-machine` freeze the clock, `moto` fakes AWS services, `fakeredis` fakes Redis.

In the interview

"I prefer dependency injection so I can pass a fake. At external boundaries I use `unittest.mock` with autospec: `return_value` to stub, `side_effect` for errors or sequences, and `assert_called_once_with` only when the interaction itself matters. With `patch` I patch where the name is looked up."

- *Mock vs stub?* A stub supplies answers; a mock also verifies how it was called.
- *Why autospec?* So the mock fails on methods or signatures the real object doesn't have.

Self-check

1. `side_effect = [1, 2]` — what do three calls do? → Return 1, then 2, then raise `StopIteration`.
2. `shop.py` has `from requests import get`. What do you patch? → `"shop.get"`.

3. What's a fake? → A simplified working implementation, e.g. an in-memory repository.

TL;DR

Doubles replace slow/external dependencies: dummy, stub, spy, mock, fake. `Mock` + `return_value` / `side_effect` + `assert_called_*`; `patch` where used; autospec to stay honest; mock only at boundaries.

15.6 Coverage

What it is

Code coverage measures which lines (and branches) of your code ran during the tests. `coverage.py` is the standard tool; `pytest-cov` integrates it with pytest.

- **Line (statement) coverage** — was each line executed?
- **Branch coverage** — did each `if` go both ways? A line can run while one side of its `if` is never taken.

How it works

```
pytest --cov=inventory --cov-branch --cov-report=term-missing
# or without the plugin:
coverage run --branch -m pytest && coverage report -m && coverage html   # htmlcov/index.html
```

Output:

```
Name           Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------
inventory.py      48      7     10      2    84%   48, 81-86
-----------------------------------------------------------
TOTAL             48      7     10      2    84%
```

Reading it: 48 statements, 7 never ran; 10 branch points, 2 partially taken. `Missing` lists the lines: **48** is `raise ValueError("qty must be positive")` in `Inventory.add` — no test ever passes a non-positive quantity, so that validation is untested. **81-86** is the `if __name__ == "__main__":` demo block (fine to exclude with a `# pragma: no cover` comment or config).

Trade-offs / how to use the number

- Coverage shows what is **not** tested. It does **not** show that tested code is **correct** — a test that runs a line without asserting anything still "covers" it.
- Many teams require a threshold in CI (`--cov-fail-under=85`). 100% is rarely worth the effort except in small critical libraries; aim for high coverage of business logic and error paths.
- Watch the *missing* list during code review more than the percentage.
- **Mutation testing** (e.g. `mutmut`) goes further: it changes your code (`<` to `<=`) and checks that some test fails.

TL;DR

`pytest --cov=pkg --cov-branch --cov-report=term-missing`. Use it to find untested lines, not as proof of quality.

15.7 doctest — examples that are tests

What it is

`doctest` finds text in docstrings that looks like an interactive session (`>>> expr` followed by the expected output) and checks that running it produces that output.

```
def slugify(name: str) -> str:
    """Lower-case, keep letters/digits, join words with '-'.

    >>> slugify("Hello, World!")
    'hello-world'
    """
```

How it works

- `python -m doctest inventory.py -v`, or inside the module `doctest.testmod()`, or with pytest: `pytest --doctest-modules`.
- Output is compared as **text**, so it must match `repr` exactly.

Output of `python inventory.py` and `pytest --doctest-modules inventory.py`:

```
stock: {'mug': 3} | units: 3
doctest: TestResults(failed=0, attempted=3)
..                                                                       [100%]
2 passed in 0.07s
```

(3 examples in 2 docstrings; pytest counts each docstring as one test.)

Trade-offs / when not to use it

Great for keeping **documentation examples correct**. Poor for real test suites: text comparison is brittle (dict order, float digits, memory addresses in reprs), and there are no fixtures or parametrization. Use doctests for docs, pytest for tests.

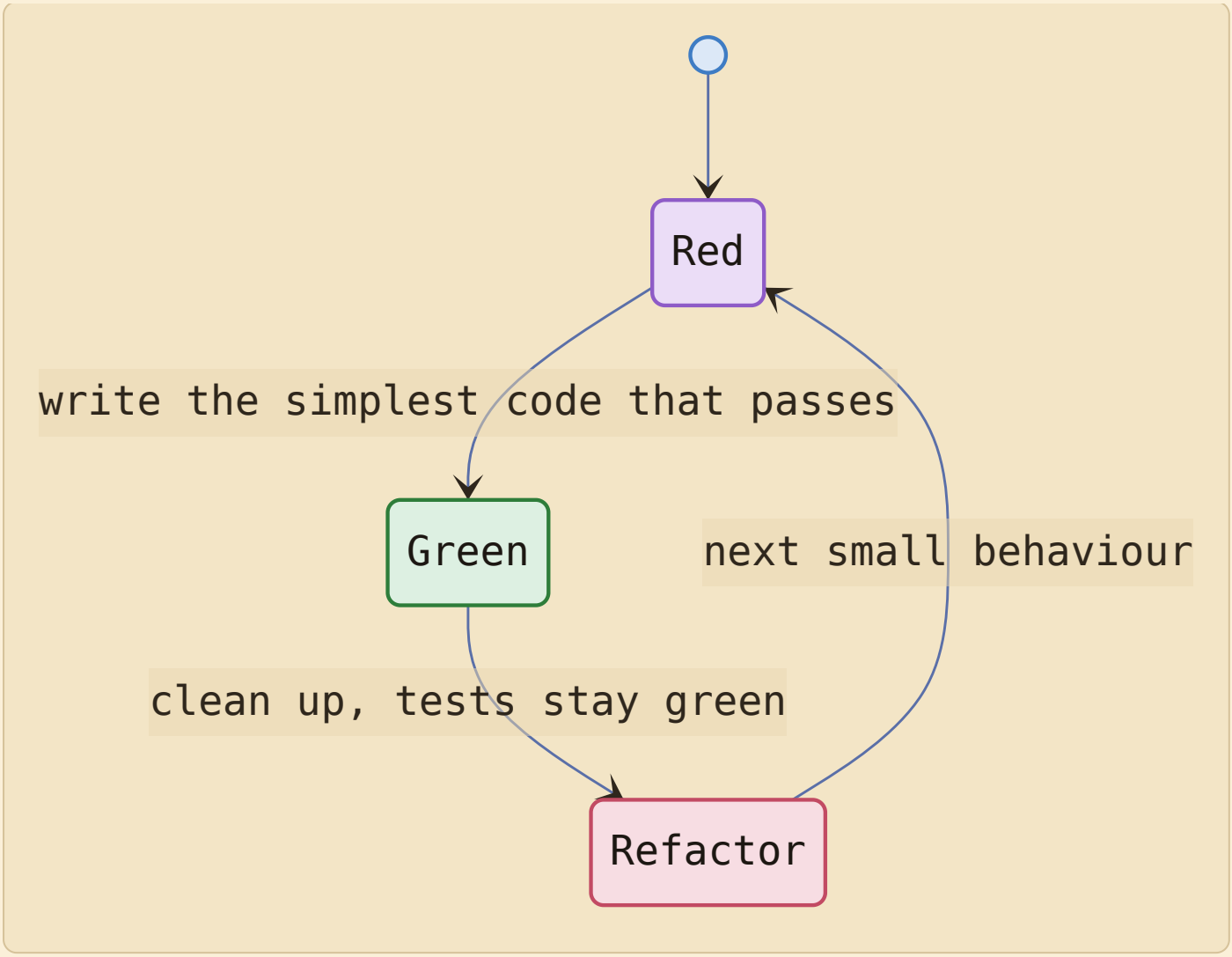
TL;DR

Docstring examples that run. Keep them short; `pytest --doctest-modules` runs them with the rest.

15.8 The TDD loop (red, green, refactor)

What it is

Test-driven development (popularised by Kent Beck) is writing the test **before** the code, in tiny cycles:



1. **Red** — write a small failing test for the next behaviour. Run it; see it fail *for the right reason*.
2. **Green** — write the minimum code to pass.
3. **Refactor** — improve names and structure; the tests guard you.

How it works — a real cycle

Feature: `Inventory.low_stock(threshold)` → SKUs whose quantity is **at or below** the threshold, sorted.

Red. Tests first (`test_tdd_low_stock.py`):

```
def test_low_stock_lists_skus_below_threshold(inv):      # inv: mug=5, pen=10
    assert inv.low_stock(threshold=6) == ["mug"]

def test_low_stock_sorted_and_inclusive(inv):
    inv.add("cap", 1)
    assert inv.low_stock(threshold=10) == ["cap", "mug", "pen"]
```

Run → both fail because the method doesn't exist yet:

```
>       assert inv.low_stock(threshold=6) == ["mug"]
               ^^^^^^^^^^^^^
E       AttributeError: 'Inventory' object has no attribute 'low_stock'
...
2 failed in 0.07s
```

Green, first try. A quick implementation `return [sku for sku, q in self.stock.items() if q < threshold]` passes the first test but not the second. pytest's assertion rewriting shows exactly why:

```
>       assert inv.low_stock(threshold=10) == ["cap", "mug", "pen"]
E       AssertionError: assert ['mug', 'cap'] == ['cap', 'mug', 'pen']
E
E         At index 0 diff: 'mug' != 'cap'
E         Right contains one more item: 'pen'
E         Use -v to get more diff
1 failed, 1 passed in 0.07s
```

Two bugs are visible: the order isn't sorted, and `pen` (exactly 10) is missing, so the comparison must be `<=`.

Green.

```python
def low_stock(self, threshold: int) -> list[str]:
    """SKUs with quantity <= threshold, alphabetically."""
    return sorted(sku for sku, qty in self.stock.items() if qty <= threshold)
```

Whole suite: `31 passed, 1 skipped, 1 xfailed`.

Refactor. Names are clear already; nothing else to do — move to the next behaviour.

Trade-offs

Pros	Cons
Every line has a reason (a test)	Slower start; feels unnatural at first
Forces small, testable design	Hard when exploring an unknown problem/API (spike first, then test)
Instant regression safety	Can over-fit tests to one design if cycles are too big

Many teams use TDD for bug fixes at least: **first write a test that reproduces the bug**, see it fail, then fix — the bug can never silently return.

TL;DR

Red (failing test) → Green (simplest fix) → Refactor. At minimum, reproduce every bug with a failing test before fixing it.

15.9 Property-based testing with Hypothesis (brief)

What it is

In an **example-based** test you pick inputs. In a **property-based** test you state a rule that must hold for *all* inputs of some kind, and a library generates hundreds of inputs — including nasty ones (empty strings, huge numbers, Unicode, NaN) — trying to break it. **Hypothesis** is Python's main library for this (the idea comes from Haskell's QuickCheck).

When it finds a failing input it **shrinks** it to the smallest, simplest failing example, and stores it in a local database (`.hypothesis/`) so it's retried first next time.

How it works

```
from hypothesis import given, settings, strategies as st

@given(st.text())
def test_slugify_is_idempotent(s):             # applying it twice changes nothing
    assert slugify(slugify(s)) == slugify(s)

@given(st.text())
def test_slug_has_only_safe_chars(s):
    assert all(c.isascii() and (c.isalnum() or c == "-") for c in slugify(s))

@given(price=st.floats(0, 1e6), pct=st.floats(0, 100))
def test_discount_never_increases_price(price, pct):
    assert apply_discount(price, pct) <= round(price, 2)

@settings(max_examples=200)
@given(st.lists(st.tuples(st.sampled_from(["add", "remove"]), st.integers(1, 20))))
def test_stock_never_negative(ops):            # random sequences of operations
    inv = Inventory()
    for op, qty in ops:
        try:
            getattr(inv, op)("mug", qty)
        except OutOfStock:
            pass
        assert inv.stock.get("mug", 0) >= 0
```

All four pass on our `slugify`. To see shrinking, run the same "safe characters" property against a naive version `name.strip().lower().replace(" ", "-")`:

```
E       Failing test case: test_slug_has_only_safe_chars(
E           s=':',
E       )
```

Out of all the random strings it tried, Hypothesis reduced the failure to the single character `':'`.

Common properties to test: **round-trip** (`load(save(x)) == x`, `decode(encode(x)) == x`), **idempotence** (`f(f(x)) == f(x)`), **invariants** (stock never negative, output sorted, length preserved), **oracle** (fast version equals a slow obviously-correct version).

When not to use it

When you can't state a general rule, or when inputs are expensive to create. Use it alongside example tests, not instead of them.

TL;DR

State properties, let Hypothesis generate and shrink counter-examples. Great for parsers, serializers and invariants.

15.10 Reading tracebacks

What it is

A **traceback** is Python's report of an uncaught exception: the chain of function calls (the **call stack**) that led to the error, oldest call at the top, the failing line at the bottom.

How it works — read it bottom-up

1. **Last line first**: the exception type and message — `ValueError: invalid literal for int() with base 10: 'two'`.
2. **The frame just above it**: the file, line and function where it happened. Since 3.11, `~~~^^^` markers point at the exact sub-expression (PEP 657).
3. **Walk upward** until you reach *your* code (skip library frames) — often the bug is where your code passed bad data in.
4. **Chained exceptions**: "The above exception was the direct cause of the following exception" means the lower exception was raised with `raise NewError(...) from original`. "During handling of the above exception, another exception occurred" means a new error happened *inside* an `except` block (implicit chaining).

From `debugging_demo.py` (a CSV-like line with `"two"` as the quantity):

```
Traceback (most recent call last):
  File "debugging_demo.py", line 19, in load_orders
    orders.append(parse_order(line))
                  ~~~~~~~~~~~^^^^^^
  File "debugging_demo.py", line 12, in parse_order
    return {"sku": sku, "qty": int(qty), "price": float(price)}
                               ~~~^^^^^
ValueError: invalid literal for int() with base 10: 'two'

The above exception was the direct cause of the following exception:

Traceback (most recent call last):
  File "debugging_demo.py", line 35, in <module>
    load_orders(["mug,2,4.50", "pen,two,1.20"])
    ~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "debugging_demo.py", line 21, in load_orders
    raise RuntimeError(f"bad order on line {n}: {line!r}") from e
RuntimeError: bad order on line 2: 'pen,two,1.20'
```

The final error tells *which input* was bad (line 2); the cause above it tells *what* was wrong with it (`int('two')`). Good error handling (`raise ... from e` with context) makes tracebacks like this.

Common exception → usual cause

Exception	Usual cause
`NameError`	typo, or used before assignment
`AttributeError: 'NoneType' object has no attribute ...`	a function returned `None` (forgot `return`, or an in-place method like `list.sort()`)
`TypeError: ... takes 2 positional arguments but 3 were given`	forgot `self`, or wrong call
`KeyError` / `IndexError`	missing dict key / off-by-one
`UnboundLocalError`	assigning to a name inside a function makes it local (use `nonlocal`/`global` or rename)
`ModuleNotFoundError`	wrong virtual environment, or a local file shadows the module (e.g. your own `random.py`)
`RecursionError`	missing or unreachable base case

TL;DR

Read bottom-up: message, failing line, then up to your code. "Direct cause" = explicit `from`; "during handling" = error inside `except`.

15.11 Debugging with `pdb` and `breakpoint()`

What it is

`pdb` is Python's built-in interactive **debugger**: it pauses the program at a line and lets you inspect variables, step line by line, and move up and down the call stack. `breakpoint()` (3.7+, PEP 553) is the built-in way to pause there — it calls `pdb.set_trace()` by default.

How it works — ways in

Way	When
Put `breakpoint()` in the code	pause at a specific spot
`python -m pdb script.py`	start the whole script under the debugger
`pdb.pm()` after a crash (in the REPL) or `python -m pdb` (it enters post-mortem on an uncaught exception)	inspect the frame where it died
`pytest --pdb`	debugger at the failing assertion
`pytest --trace`	debugger at the start of each test
IDE (VS Code, PyCharm) breakpoints	same ideas, with a GUI

The environment variable `PYTHONBREAKPOINT` controls what `breakpoint()` does: `PYTHONBREAKPOINT=0` disables all of them (safety net in CI), `PYTHONBREAKPOINT=ipdb.set_trace` switches to another debugger.

The commands you need

Command	Does
`l` / `ll`	list code around the current line / the whole function
`n` (next)	run the current line, stepping *over* calls
`s` (step)	step *into* a call
`r` (return)	run until the current function returns
`c` (continue)	run until the next breakpoint
`unt N` (until)	run until line N
`p expr` / `pp expr`	print / pretty-print an expression
`w` (where)	show the call stack
`u` / `d`	move up / down the stack to inspect callers
`b file:line` / `b func` / `b line, cond`	set a (conditional) breakpoint
`cl N`	clear breakpoint N
`display expr`	show `expr` each time execution stops if it changed
`interact`	a full Python REPL in the current scope
`q`	quit

Tip: a bare expression also works, but if a variable is named like a command (`n`, `c`, `l`…), use `p n`.

A real session (scripted, so it can be shown)

`debugging_demo.py` drives `pdb` with a list of commands instead of the keyboard. The function under investigation crashes on an empty list:

```python
def average_value(orders: list[dict]) -> float:
    total = 0.0
    for o in orders:
        total += o["qty"] * o["price"]
    return total / len(orders)          # bug when orders is empty
```

Commands: `b average_value`, `c`, `p orders`, `n`, `n`, `p total, len(orders)`, `c`. Transcript:

```
(Pdb) Breakpoint 1 at debugging_demo.py:26
(Pdb) > debugging_demo.py(26)average_value()
-> total = 0.0
(Pdb) []
(Pdb) > debugging_demo.py(27)average_value()
-> for o in orders:
(Pdb) > debugging_demo.py(29)average_value()
-> return total / len(orders)          # bug when orders is empty
(Pdb) (0.0, 0)
(Pdb)
then it raised: ZeroDivisionError('division by zero')
post-mortem locals: {'orders': [], 'total': 0.0}
breakpoint() skipped because PYTHONBREAKPOINT=0
```

Stepping shows the loop body is skipped and `len(orders)` is 0 at the return — the fix is to handle the empty case (`if not orders: return 0.0` or raise a clear error). The "post-mortem" line shows the same information recovered from the traceback's innermost frame after the crash, which is what `pdb.pm()` gives you interactively.

A debugging method (not just a tool)

1. **Reproduce** reliably — ideally as a failing test.
2. **Read** the traceback and the relevant code; state a hypothesis.
3. **Observe**: `breakpoint()`, logging, or `p` in pdb to check the hypothesis.
4. **Narrow down**: halve the input, comment out half the code, or use `git bisect` to find the commit that introduced the bug.
5. **Fix** the cause, not the symptom; keep the test so it can't regress.
6. Explaining the code line by line to someone (or a rubber duck) surprisingly often reveals the bug.

Common mistakes

- Committing a `breakpoint()` (CI hangs waiting for input — set `PYTHONBREAKPOINT=0` there, and use a linter rule such as Ruff's `T100`).
- Changing several things at once while debugging.

TL;DR

`breakpoint()` → `n`/`s`/`c`/`p`/`w`/`u`/`d`. `python -m pdb`, `pytest --pdb`, `pdb.pm()` for post-mortem. Reproduce, hypothesise, observe, narrow, fix, keep the test.

15.12 Logging vs `print` for debugging

What it is

`print` debugging is fine for a five-minute investigation. For anything that stays in the code, use **logging** (Chapter 14): it has levels, timestamps, module names and configurable destinations, and can be turned up to DEBUG only where you need it.

	print	logging
Levels / filtering	no	DEBUG…CRITICAL, per module
Timestamp, module, line	manual	via formatter
Destination	stdout	console, files, syslog, log services
Safe to leave in code	no	yes (DEBUG off in production)
In tests	captured by pytest (`-s` shows)	`caplog` fixture lets tests assert on logs

```
log = logging.getLogger(__name__)
log.debug("pricing order=%s rate=%s", order_id, rate)    # visible only at DEBUG
```

Tips: temporarily raise one module's level (`logging.getLogger("shop.pricing").setLevel(logging.DEBUG)`); f-string `=` specifier for quick prints — `print(f"{total=}")` prints `total=0.0`; `python -X dev` enables extra runtime checks and warnings; `faulthandler` (`python -X faulthandler`) dumps tracebacks on hard crashes or hangs.

TL;DR

`print` for a quick look, then delete it; logging for anything that stays. Test logs with `caplog`.

15.13 Common bugs checklist

Run through this list when something "can't possibly be wrong":

#	Bug	Example	Fix
1	Mutable default argument	`def f(items=[])` shares one list across calls	`items=None` then `items = [] if items is None else items`
2	Late-binding closures	`[lambda: i for i in range(3)]` all return 2	`lambda i=i: i`
3	Aliasing / shallow copies	`grid = [[0]*3]*3` — one row shared three times	`[[0]*3 for _ in range(3)]`, `copy.deepcopy`
4	Modifying a list while iterating	removing items inside `for x in xs` skips items	iterate over a copy or build a new list
5	`is` vs `==`	`x is 1000` depends on object identity	`==` for values; `is` only for `None`, sentinels
6	Float equality	`0.1 + 0.2 == 0.3` is False	`math.isclose`, `pytest.approx`, `Decimal` for money
7	Integer vs float division	`7 / 2 == 3.5`, `7 // 2 == 3`, `-7 // 2 == -4`	pick the right operator
8	Methods returning `None`	`xs = xs.sort()` makes `xs` None	`xs.sort()` or `xs = sorted(xs)`
9	Swallowed exceptions	`except Exception: pass` hides the real error	catch specific exceptions, log, re-raise
10	Off-by-one	`range(1, n)` excludes `n`	test the boundaries
11	Truthiness	`if count:` is False for `0`	`if count is not None:`
12	Naive datetimes / time zones	comparing local and UTC times	aware UTC datetimes (Chapter 14)
13	Unclosed resources	files/connections left open	`with` statements
14	Wrong environment	`ModuleNotFoundError` though "it's installed"	check `sys.executable`, activate the venv, `python -m pip`
15	Module name shadowing	your `random.py` or `test.py` hides the stdlib	rename the file
16	Global/shared state between tests	tests pass alone, fail together	fresh fixtures, no module-level mutable state
17	Encoding	`UnicodeDecodeError` reading files	`open(p, encoding="utf-8")`
18	Unseeded randomness / real clock in tests	flaky tests	seed, inject the clock, freeze time

TL;DR

Most Python bugs are a handful of classics: mutable defaults, aliasing, `None` returns, float equality, swallowed exceptions, wrong venv. Check the list before a long debugging session.

15.14 Chapter self-check

1. Name the three parts of a well-shaped test. → Arrange, Act, Assert.
2. Why do most tests belong at the bottom of the pyramid? → Unit tests are fast, precise and cheap.
3. How does pytest show the values in a failing `assert`? → Assertion rewriting.
4. How do you add cleanup to a fixture? → Code after `yield`.
5. What does `tmp_path` give you? → A unique temporary directory (`Path`) per test.
6. `xfail(strict=True)` and the test passes — result? → Reported as a failure (XPASS strict), so you notice the fix.
7. Stub vs mock vs fake? → Canned answers / verified interactions / working simplified implementation.
8. Why `create_autospec`? → Mocks reject attributes and signatures the real object lacks.
9. What does 100% line coverage prove? → Only that every line ran — not that results were checked.
10. TDD cycle? → Red, green, refactor.
11. What does Hypothesis do after finding a failing input? → Shrinks it to a minimal example and remembers it.
12. How do you read a traceback? → Bottom-up: message, failing line, then up to your code.
13. How do you disable every `breakpoint()` in CI? → `PYTHONBREAKPOINT=0`.
14. `xs = xs.sort()` — what is `xs`? → `None`.

Chapter TL;DR

- **Why:** tests make change safe. Many fast unit tests, fewer integration/E2E; deterministic, isolated, behaviour-focused; run in CI.
- **unittest:** `TestCase`, `setUp`, `assertEqual`/`assertRaises`; built in.
- **pytest:** plain `assert` (with rewritten failure diffs), fixtures (`yield` teardown, scopes, `conftest.py`), `parametrize`, `tmp_path`, `monkeypatch`, `capsys`, `caplog`, marks (`skip`, `skipif`, `xfail`, custom + `-m`), `-k`, `-x`, `--lf`, `--pdb`.
- **Doubles:** dummy, stub, spy, mock, fake. `Mock` + `return_value`/`side_effect`/`assert_called_once_with`; `patch` where the name is used; autospec; mock only at boundaries; prefer dependency injection.
- **Coverage:** `pytest --cov --cov-branch --cov-report=term-missing` finds untested code; it doesn't prove correctness.
- **TDD:** red, green, refactor; always reproduce a bug with a failing test first.
- **doctest** keeps doc examples true; **Hypothesis** checks properties and shrinks counter-examples.
- **Debugging:** read tracebacks bottom-up; `breakpoint()`/`pdb` (`n s c p w u d b`), `pytest --pdb`, post-mortem; reproduce → hypothesise → observe → narrow → fix → keep the test.
- **Logging over print** for anything that stays; and check the **common bugs list** first.